Final Project Report

Metro Ticket Generating System in ServiceNow

ServiceNow Global Certification Program

1. INTRODUCTION

1.1 Project Overview

Metro Ticket Generating System is a ServiceNow Service Catalog solution designed to digitize metro ticket booking. The commuter selects source and destination stations, chooses a single or return journey, enters the number of passengers, reviews the calculated fare and receives a QR-based digital ticket.

1.2 Purpose

The purpose is to replace repetitive manual ticketing steps with a structured digital flow that captures consistent station and journey information, calculates route fares automatically and provides a digital ticket output.

2. IDEATION PHASE

2.1 Problem Statement

User	I'm trying to	But	Because
Metro commuter	Buy a correct ticket quickly	Queues and manual steps	Ticketing is not fully digitized
Metro commuter	Know the fare before checkout	Fare can require manual calculation	Route rules are not automated

2.2 Empathy Map Canvas

Commuters want speed, certainty and a ticket they can access digitally. The solution therefore prioritizes guided inputs, visible fare calculation and QR output.

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3. PROBLEM — SOLUTION FIT

The identified problem is slow, fragmented and manual ticket preparation. The proposed ServiceNow solution directly addresses this by centralizing route selection, fare logic, passenger scaling and digital ticket generation in one catalog item.

4. PROPOSED SOLUTION

Parameter	Description
Problem	Commuters need a faster and more consistent way to select a route and obtain a metro ticket.
Idea / Solution	ServiceNow catalog item “Book A Metro Ticket” with station references, journey choices, passenger count, fare calculation, payment selection and QR generation.
Novelty	Combines configurable catalog variables with route-specific JavaScript fare logic and QR ticket output.
Business Value	Reduces manual calculations, improves consistency and supports a digital-first commuter journey.

5. STAKEHOLDERS

Stakeholder	Role
Passengers	Book tickets and use QR output.
Station Managers	Maintain awareness of station data and operational needs.
Metro Operations Team	Use transaction information for service planning.
IT Administrators	Maintain ServiceNow configuration, scripts and access controls.

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6. FUNCTIONAL REQUIREMENTS

FR	Requirement	Implementation
FR-1	Station selection	Reference variables use Metro Station’s Details table.
FR-2	Journey selection	Single Journey and Return Journey choices.
FR-3	Passenger count	Select Box captures number of passengers.
FR-4	Fare calculation	Catalog Client Script calculates configured route fares.
FR-5	Payment selection	Payment-related catalog variables support checkout flow.
FR-6	QR ticket	onSubmit opens QR widget using generated ticket URL.

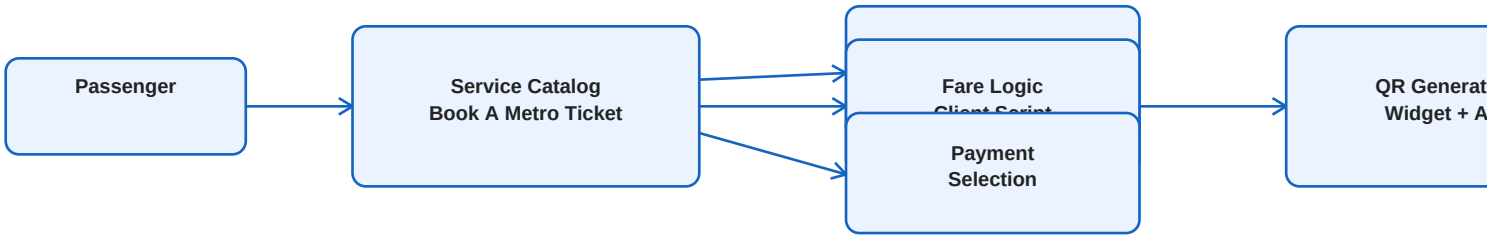
7. NON-FUNCTIONAL REQUIREMENTS

- Usability — simple guided catalog form.
- Performance — client-side fare calculation is immediate for configured routes.
- Maintainability — station records are separated into a custom station table.
- Security — production payment processing should use compliant gateway integration and appropriate ACLs.
- Scalability — additional stations and routes can be added to the data model and fare logic.

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8. TECHNICAL DESIGN



8.1 ServiceNow Components

Component	Purpose
Service Catalog Item	Book A Metro Ticket user interface
Catalog Variables	Source, destination, journey, passengers, amount and payment
Custom Table	Metro Station’s Details stores station master data
Catalog Client Script	Calculates fares and controls checkout/QR behavior
Service Portal	Presents the booking experience
QR Widget / QR API	Displays the generated digital ticket QR

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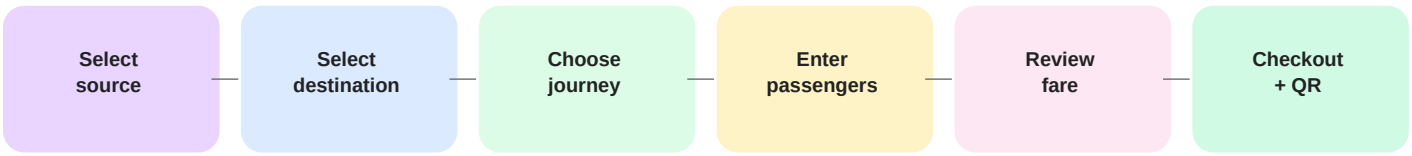
9. IMPLEMENTED FARE LOGIC

Source	Destination	Single	Return
Ameerpet	Madhapur	■30	■60
Ameerpet	LB Nagar	■50	■100
Ameerpet	Kukatpally	■30	■60
Ameerpet	Jubilee Hills	■20	■40

9.1 Fare Calculation Behaviour

The catalog reference variables return station sys_ids. The fare script maps the actual PDI station sys_ids to the configured route fares. Single journey uses the single fare per passenger; return journey uses the return fare per passenger. The total is scaled by passenger count.

10. EXECUTION ROADMAP

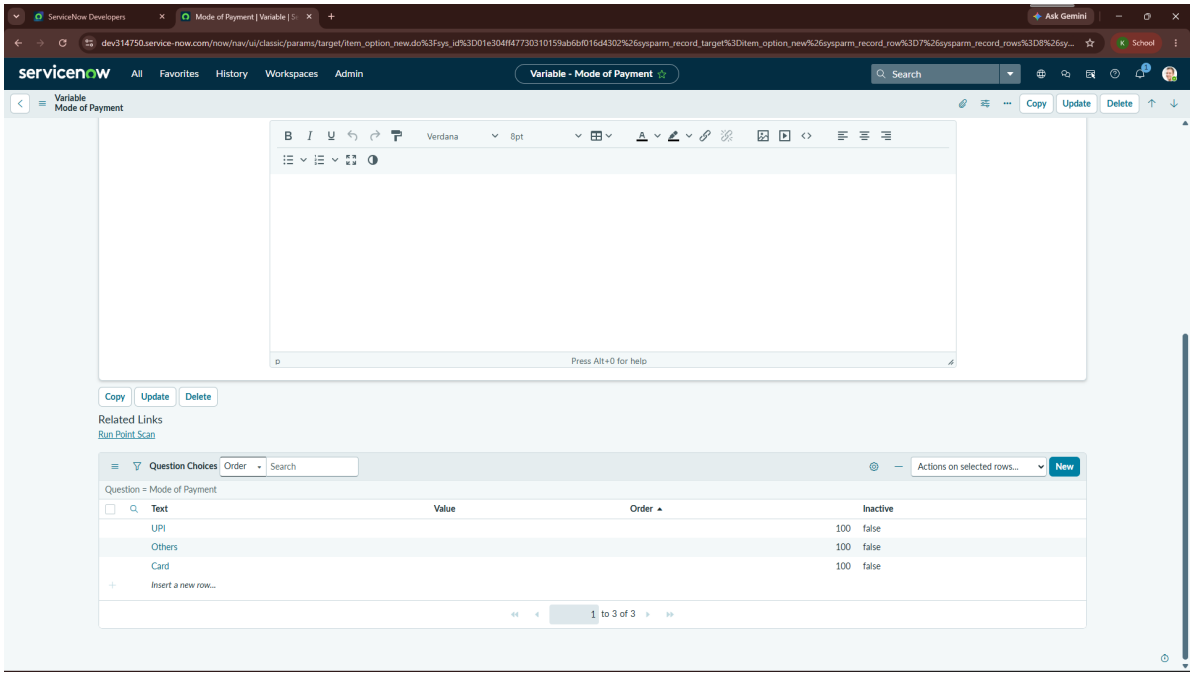


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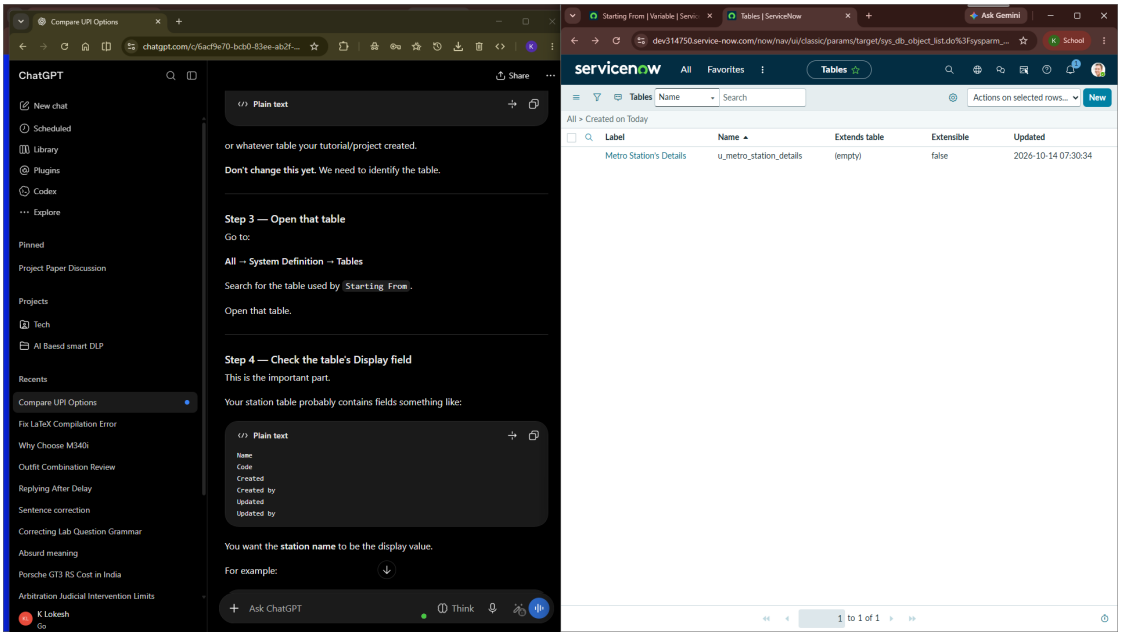
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11. SERVICE CATALOG IMPLEMENTATION

The actual PDI configuration includes the “Book A Metro Ticket” catalog item, reference station variables and passenger/journey inputs.



11.1 Station Master Data

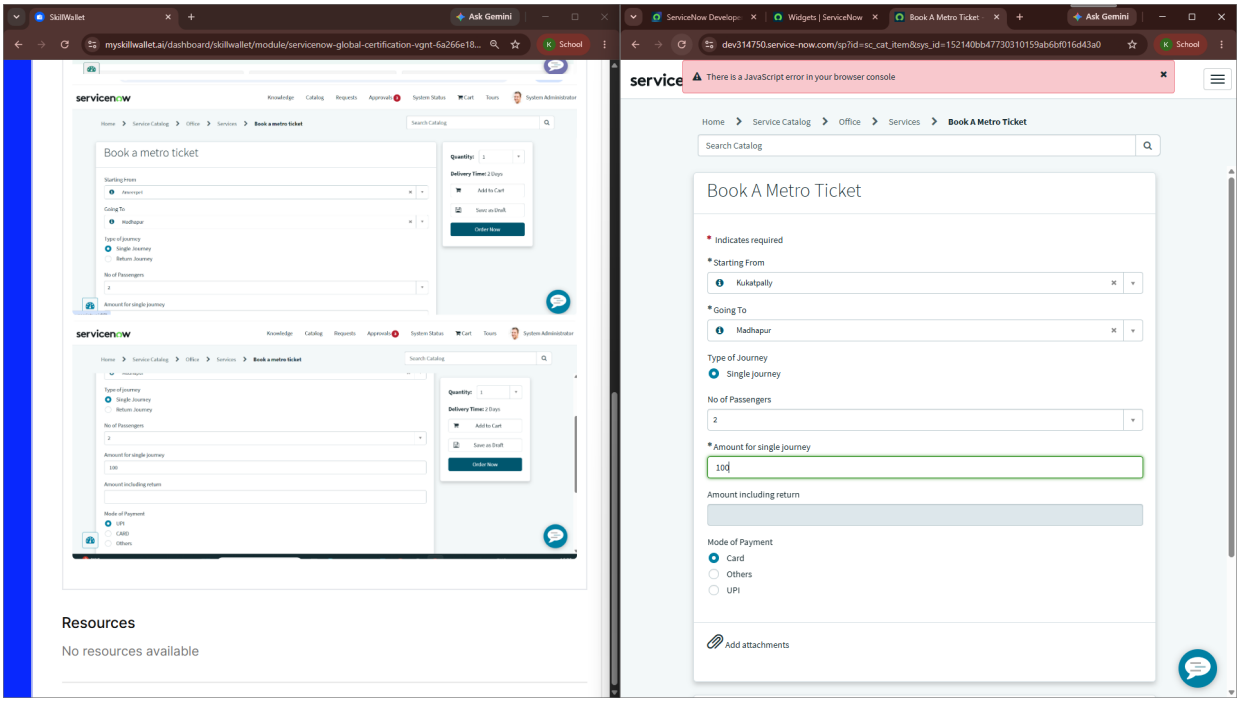


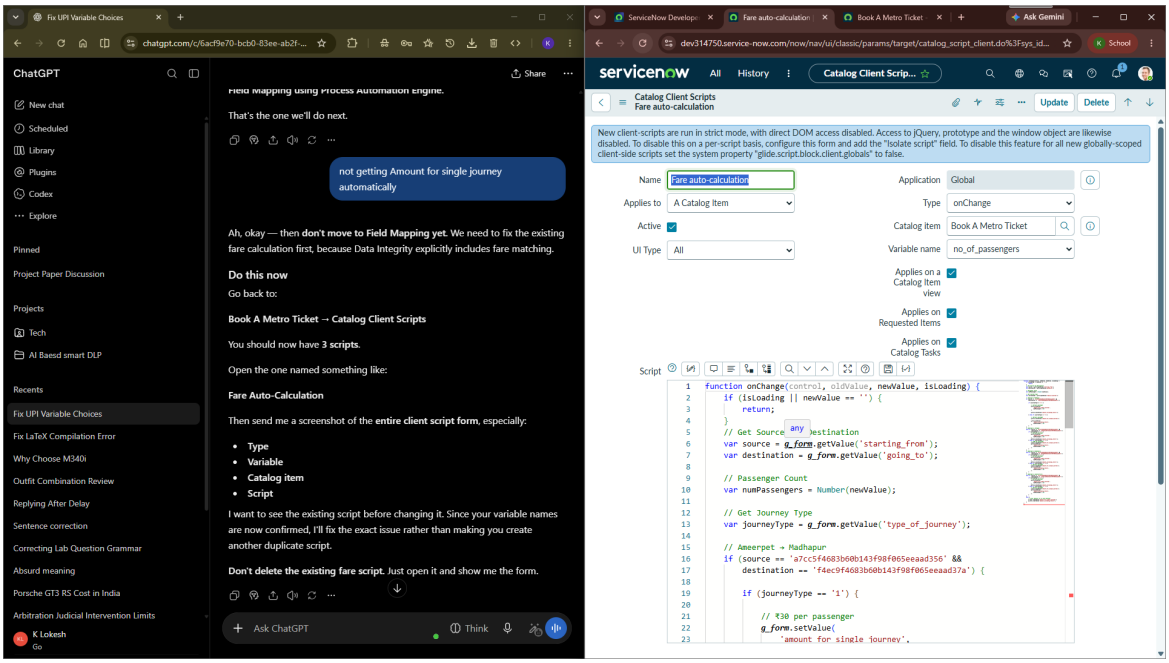
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12. USER INTERFACE & FARE CALCULATION

The Service Portal booking form presents the commuter-facing catalog experience. Route selection and journey type drive the calculated amount.



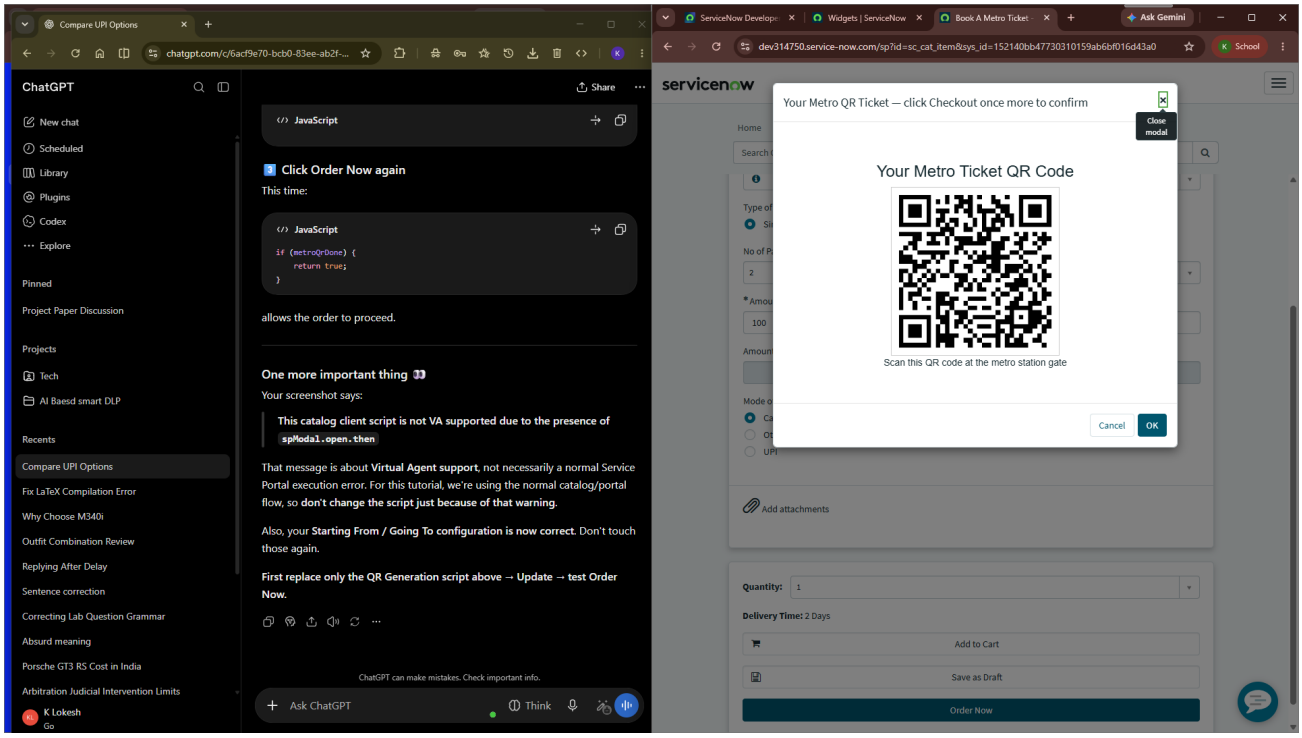


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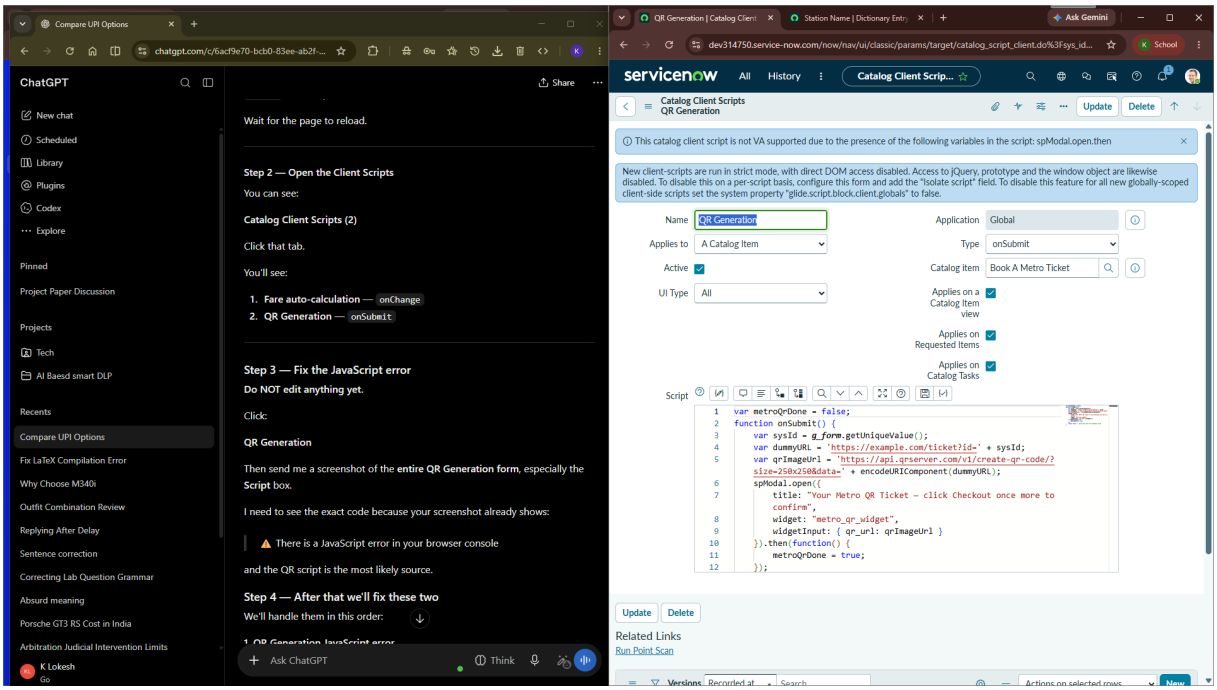
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13. QR TICKET GENERATION

After checkout, the project opens a QR ticket widget. The QR is generated from a ticket URL and presented as a digital output that can be scanned or retained on the commuter's device.



13.1 QR Client Script



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14. TESTING & VALIDATION

Test	Expected	Result	Status
Ameerpet → Madhapur, Single	■30/passenger	■30/passenger	PASS
Ameerpet → LB Nagar, Single	■50/passenger	■50/passenger	PASS
Ameerpet → Jubilee Hills, Single	■20/passenger	■20/passenger	PASS
Return journey	Return fare applied	Return fare applied	PASS
Passenger scaling	Total scales by count	Validated	PASS
QR generation	QR modal appears	Validated	PASS
Payment mode	Payment choice available	Validated	PASS

15. FUTURE ENHANCEMENTS

Future versions can add a production payment gateway, richer ticket records, dynamic route/fare tables instead of hardcoded mappings, QR validation at entry gates, commuter notifications, reporting dashboards and AI-assisted travel recommendations.

16. CONCLUSION

The Metro Ticket Generating System demonstrates a complete ServiceNow catalog-based digital ticketing flow from station selection to fare calculation and QR output. The implementation is suitable as a PDI MVP and provides a clear foundation for a production-grade metro ticketing platform.